COUNTRY PROFILES

US Nuclear Fuel Cycle

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- The USA has nuclear fuel production capacity insufficient for domestic needs.
- Currently, almost all the uranium used in US commercial reactors is imported. After reaching a peak in 1980, domestic mining now accounts for about 5% of the fuel used in US reactors.
- Between 1977 and 2005, government policy did not allow reprocessing of used fuel for commercial reactors.

Privacy - Terms

Uranium resources and mining

The USA ranks 14th in the world for known uranium resources in the category up to $130/kgU ($50/lb U_3O_8), with 59,400 tU (reasonably assured resources, 2021), about 1% of world total.

In the 1950s, the USA had a great deal of uranium mining, promoted by federal subsidies. Peak production since 1970 was 16,800 tU in 1980, when there were over 250 mines in operation. This abruptly dropped to 50 in 1984 when 5,700 tU was produced, and then there was a steady decline to 2003, by which time there were only two small operations producing a total of under 1000 tU/yr, or about 5% of the uranium consumed by US nuclear plants. So, for the first step in the nuclear fuel cycle, the US must rely on imports of uranium from countries such as Canada and Australia. For some time the country's reactors also used downblended weapons-grade uranium from Russia (see section on Military surplus and other government stocks below).

As the price of uranium increased from historic lows, there was a revival in exploration and plans to reopen old mines. Exploration expenditure increased over 2007-08 to $50.3 million, but then dropped again with uranium prices. Since 2022, exploration activity has increased again, and decisions have been made to restart many mines.

Most US production has been from New Mexico and Wyoming. Some 40% of resources are in New Mexico and amenable to in-situ leaching (ISL). Resources on the western Colorado Plateau require conventional mining and milling, as does high-grade breccia pipe mineralization in northwest Arizona. Smaller resources are in Utah, Nebraska and Texas. Production potential is about 50% underground mining, 26% open pit and 24% ISL.

US uranium production, tonnes

	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
U_3O_8	1866	2061	1774	1685	1922	1814	1882	2113	2263	1481	1327
U	1583	1748	1504	1429	1630	1538	1596	1792	1919	1256	1125

In mid-2009, the Nuclear Regulatory Commission (NRC) issued a generic environmental impact statement (EIS) on ISL (or in situ recovery, ISR) mining in western USA. This streamlined the requirement for a supplementary EIS for each new mine.

Details of US uranium mining are in US Nuclear Fuel Cycle Appendix 1: US Uranium Mining and Exploration.

The US Energy Information Administration's (EIA's) 2023 Uranium Marketing Annual Report said that 23,400 tonnes of U_3O_8e (equivalent) was purchased by US utilities in 2023, and 95% of this was from foreign sources. Only 1080 tonnes of US origin U_3O_8e (equivalent) was delivered in 2023. Canada provided 5970 t (25%), Kazakhstan 4820 t (21%), Australia 4810 t (21%), Russia 2740 t (12%), Uzbekistan 2220 t (9%), Namibia 700 t (3%), Niger 640 t (3%). Other sources provided about 960 t. During 2023, about 15% was purchased under spot contracts and the balance under long-term contracts or other arrangements, the overall weighted average price being $43.8/lb U_3O_8 ($113.9/kgU). Total US commercial inventory was 68,900 t U_3O_8e (58,500 tU).

In July 2018, the US Department of Commerce launched a Section 232 investigation* into whether the USA's reliance on uranium imports poses a threat to national security. Prior to that, the US Department of the Interior determined uranium to be a critical mineral. In July 2019, the petition was rejected.

* Under Section 232 of Trade Expansion Act of 1962 (as amended), investigations may be carried out to determine the effects of imports on the USA's national security and give the President the ability to address any threats to national security by restricting imports through tariffs.

In September 2022 as part of the $47 billion in supplemental funding request in the Biden administration's short-term spending bill, $1.5 billion in funds was requested "for the acquisition and distribution of low-enriched uranium (LEU) and high-assay LEU (HALEU)... to reduce the reliance of the United States and friendly foreign countries on nuclear fuels from the Russian Federation and other insecure sources of LEU and HALEU."

At the end of September 2022, in an amendment of the National Defense Authorization Act, a bipartisan group of US Senators introduced a ban on Russian nuclear fuel imports. The amendment would see the prioritization of domestic production of LEU and the acceleration of efforts to establish a domestic HALEU capability.

In October 2023 the Nuclear Fuel Security Act was approved by the House of Representatives' Energy and Commerce Subcommittee on Energy, Climate, and Grid Security. The Act aims to establish and expand critical US nuclear fuel programmes to boost domestic uranium mining, production, enrichment, and conversion capacity.

Prohibiting Russian Uranium Imports Act

In May 2024 President Joe Biden signed the Prohibiting Russian Uranium Imports Act. The legislation bans the import of Russian-produced unirradiated LEU as well as unirradiated LEU that has been swapped for banned uranium.

The bill came into effect on 11 August 2024, and will last until the end of 2040. Waivers may be granted to allow the import of limited amounts of LEU until 1 January 2028 if one of the following criteria is met:

- The import is necessary to maintain the viability of a US nuclear energy company that is critical to the US nuclear energy fuel supply chain.
- The import is intended to support an existing arrangement to provide fuel for a nuclear power plant in another country and thus minimize the likelihood of that country seeking a non-US fuel supplier.

The US Department of Energy (DOE) has committed to acquiring LEU generated from new sources of US domestic enrichment capacity. These can include new enrichment facilities or projects that expand the capacity of existing enrichment facilities. The DOE intends to sell the LEU to utilities operating US reactors to support clean energy generation and sever reliance on Russian imports.

In July 2024 Centrus received a waiver to allow it to import LEU from Russia for delivery to US customers in 2024 and 2025.

Conversion

The large Honeywell Metropolis Works plant (MTW) in southern Illinois converts uranium oxide, U_3O_8, to uranium hexafluoride, UF_6. It is the only conversion plant in the USA. The facility was built in the 1950s under government contract to meet military conversion requirements, and began providing UF_6 for civilian use in the late 1960s. Original namplate capacity was up to 15,000 tU/yr, but this was reduced to 7000 tU/yr in 2017 in light of global demand.

In November 2017 Honeywell announced that it was idling production at the plant, owing to global oversupply and a decrease in demand. The company stated that it did not expect demand to increase before 2020. In March 2020, the NRC renewed MTW's operating licence for an additional 40 years to March 2060. In April 2021 it was announced that the plant was to restart in 2023 following upgrade work.

ConverDyn, a partnership between affiliates of Honeywell and General Atomics, is the exclusive agent for conversion sales from Honeywell-MTW.

In December 2022 ConverDyn received a $14 million award for uranium conversion services from the DOE under its $75 million programme to create a domestic uranium reserve to boost energy security. The establishment of the reserve programme was recommended by a Trump administration nuclear fuel working group tasked with a strategy to revitalize the US domestic nuclear fuel cycle.

In January 2023 Western Uranium & Vanadium Corp announced that a new mineral processing facility is to be constructed at the Green River Industrial Park in Utah. The facility is expected to be licensed and constructed for annual production of 2 million pounds U_3O_8 (770 tU) and 6-8 million pounds of V_2O_5, with initial production expected to begin in 2026.

Enrichment

Three uranium enrichment plants – at Oak Ridge, Tennessee; Paducah, Kentucky; and Portsmouth, Ohio – were built by the federal government in the 1940s and 1950s for defence purposes and operated for about 25 years. From 1969 to 1992, the US Department of Energy (DOE) and its predecessor agencies sold some of the plants' enrichment services commercially under full-cost recovery contracts that required utility customers to pay for the plants' future decommissioning. These fees were collected once, and then further fees were levied from 1992 for decommissioning.

The USA currently has one operating enrichment plant owned by Urenco, with a capacity of 4.9 million SWU per year, compared with US requirements of about 15 million SWU/yr.

The US nuclear industry has asked the DOE to provide a source of high-assay low-enriched uranium (HALEU, up to 20% U-235) for advanced reactors. Eleven developers of such reactors estimated in 2020 that they would need a total of about 2000 tonnes of HALEU above 10% enriched by 2035. Annual requirements grow progressively to 500 tU in that period.

In October 2023 the Biden administration requested $2.2 billion from Congress to enhance domestic enrichment capabilities for LEU and HALEU.

Urenco USA (formerly National Enrichment Facility)

Urenco USA (UUSA) has a centrifuge enrichment plant at Eunice, New Mexico. It uses 6th generation Urenco technology from Europe, and was planned by the Louisiana Energy Services (LES) partnership – comprising Urenco, Exelon, Duke Power, Entergy, and Westinghouse. Construction of the $1.5 billion plant was licensed by the Nuclear Regulatory Commission (NRC) in mid-2006 when as agreed the three utilities then passed their share to Urenco, and the company is now a subsidiary of Urenco USA. Utility support for the venture – initially amounting to $3.15 billion in orders – was crucial in persuading the NRC that further US enrichment capacity was required beyond that provided and envisaged by USEC.

The NRC approved commercial operation in mid-2010, with phase 1 capacity of 1.6 million SWU/yr (21 cascades of TC-12 centrifuges) being reached in 2012. Phase 2 of 2.1 million SWU/yr was inaugurated in August 2012 with the commissioning of the first new type of centrifuge, and completion to 3.7 million SWU then licensed capacity was reached in April 2014. Phase 3 began operation in 2015, taking capacity to 4.8 million SWU/yr but with

the second part of phase 3 deferred until market conditions improve. A new TC-21 centrifuge design with 60% greater capacity is being used in phases 2 &3, and the plant is licensed to enrich to 5.5% U-235 and is capable of 6%. HALEU of 18-19% U-235 – for new designs of small modular reactors – would require a further cascade.

Earlier in November 2012 it was reported that Urenco had approached Korea Electric Power Corporation (Kepco) to invest in phase 2-3 of the plant, which would have had strategic significance for Korea in relation to its nuclear agreement with the USA. Kepco buys 2.9 million SWU/yr of enrichment services from several international sources. This did not proceed.

The plant provides one-third of US enrichment demand from 63 cascades in operation and is a major step forward in underwriting new US nuclear generating capacity and in ensuring security of fuel supply, with flexibility of operation enabling more energy input to produce more fuel from the same natural uranium feed if required.

NRC approval for a licence amendment to increase capacity to 10 million SWU/yr was granted in March 2015. This expansion would involve three phases, with three new separation building modules additional to the present two, and other plant.

Earlier in May 2014 Urenco applied to the NRC for a licence amendment to allow it to use high-assay tails (c 0.4% U-235) as feed in part of the new plant expansion applied for in 2012, and this was also approved in March 2015.

In February 2019 the company said it was planning production of high-assay LEU (HALEU), beyond 5% but under 20% enriched. Initially this would be to 6% or less, requiring no more than a licence amendment, but eventually to higher levels for small reactors. It is considering construction of a dedicated HALEU plant for this, particularly to supply a new generation of small reactors. Two US facilities already produce high-enriched-uranium – well above 20% – for naval fuel. However, being foreign-owned and using non-US technology, the Urenco USA plant would be ineligible to supply enrichment services for US defence purposes. In May 2020 the NRC approved a licence amendment for the plant to enrich to 5.5%.

In March 2023 it was announced that Urenco would supply enriched uranium to Ultra Safe Nuclear Corporation to manufacture fuel for its Micro-Modular Reactor, Urenco's first supply of enriched uranium product (EUP) for use in an advanced nuclear reactor. The EUP would be produced and supplied from Urenco USA's

enrichment facility in New Mexico. The first batch of EUP is scheduled for delivery in 2025.

In July 2023 Urenco announced it would increase capacity at its Eunice plant by 700,000 SWU/yr, setting a target for the first new cascades to be in operation in 2025.

USEC and Paducah old plant

This was the last gaseous diffusion plant still operating in the world, having been commissioned in 1952 for military use. It began providing enriched uranium for civilian reactors in the 1960s. Originally government-owned, USEC became a private sector corporation in 1998, and leased its two large enrichment plants from the DOE. In 2001, it consolidated its enrichment operations at the Paducah site in Kentucky after closing the older Portsmouth facility at Piketon, Ohio[a]. Both plants were very energy-intensive and costly to run.

The Paducah plant had a capacity of 8 million SWU/yr, compared with the 12.7 million SWU/yr required by the 104 then operational US reactors. Its final role was under a complex May 2012 agreement where USEC enriched about 9000 tonnes of DOE high-assay depleted uranium tails at Paducah to provide about 480 tonnes of low-enriched uranium for two US utilities – TVA and Energy Northwest. With other work, this will required about 5 million SWU and extended USEC's electricity supply agreement with TVA for about a year. The Paducah plant closed at the end of May 2013 after more than 60 years operation. USEC said that "We will continue to meet our customers' orders from our existing inventory, purchases from Russia under the historic Megatons to Megawatts program, and our transitional supply contract with Russia that runs through 2022."

About 115,000 tonnes of depleted uranium hexafluoride remain on the site and at Portsmouth, Ohio. In November 2013 the DOE announced that it had selected a proposal from Global Laser Enrichment to build a plant to enrich most of this (see below). In the same announcement, the DOE said it would enter negotiations with Areva (now Framatome) to process off-specification uranium hexafluoride as blend stock for domestic nuclear fuel. This would be carried out using Areva's existing nuclear fuel fabrication facility in Richland, Washington.

The DOE said that the GLE and Areva projects represented "an important next step" in planning for potential future uses and

clean-up efforts at Paducah as well as reducing the costs to the taxpayer of the clean-up operation.

Fluor had a three-year $420 million DOE contract to clean up the Paducah site from 2014, then in May 2017 DOE awarded a $1.5 billion contract to Four Rivers Nuclear Partnership LLC for the continued deactivation and remediation (D&R) of Paducah and clean-up of the 1400 ha site. Four Rivers is a CH2M-led company with partners Fluor and BWX Technologies. The D&R contract is valued at $750 million for five years, followed by three-year and two-year option periods together worth about $750 million.

USEC/Centrus and Russian HEU

USEC was the agent for supply to USA of blended-down Russian uranium from weapons stockpiles (see Megatons to Megawatts section of US Nuclear Power Policy paper and Military Warheads as a Source of Fuel). This arrangement, supplying about half US demand from 1994, finished in 2013, and involved about 5.5 million SWU/yr.

USEC in March 2011 signed a further contract with Tenex for supply of low-enriched uranium from 2013 to 2022, ramping up to about half of earlier levels from Russia, with an option to match those levels. The 2011 contract covers the supply of 21 million SWU to USEC/Centrus, worth $2.8 billion. The new supplies will come from mined uranium enriched in Russia, rather than recycled weapons. A renewed supply of Russian uranium to the USA was not unexpected, but the particular arrangement allowed for USEC's new American Centrifuge Plant (see below) failing to proceed, though most of the new supply is directed at USEC customers outside the USA. USEC said the 2011 contract was complementary to its "ongoing efforts to deploy the American Centrifuge Plant". However, it also involved a feasibility study on deploying Russian centrifuge enrichment technology in the USA. USEC/Centrus purchases the SWU as LEU and delivers a corresponding amount of natural uranium to Tenex for the uranium component of the LEU, so like its predecessor, the contract yields no net uranium for the USA. In December 2015 the 2011 agreement was modified to reflect lower enrichment demand and extended to 2026 for delivery of the balance of the contracted SWU (17 million over 2016 to 2022). In July 2024 Centrus received a waiver under the Prohibiting Russian Uranium Imports Act to allow it to import LEU from Russia for delivery to US customers in 2024 and 2025.

From 2009, Russia's Tenex signed a number of contracts with US utilities to supply enrichment services and enriched uranium product. To mid-2010 the contracts totaled some $3 billion, covering supplies to 2020. This Russian supply was in addition to that of 4.4% low-enriched uranium supplied through USEC having been blended down from Russian weapons material which supplied about half of US annual needs to 2013. (see section below on Military surplus and other government stocks and Megatons to Megawatts section of US Nuclear Power Policy paper).

USEC/Centrus American Centrifuge Plant

In April 2007, the Nuclear Regulatory Commission licensed construction and operation of USEC's American Centrifuge Plant in Piketon, Ohio. The American Centrifuge technology had been developed over many years by USEC, based on work by the Department of Energy (DOE) in 1970s and 1980s. The plant was constructed on the same Portsmouth site where the DOE's experimental plant operated in the 1980s, involving 1300 centrifuges as the culmination of a very major R&D programme. It is also the site of USEC's large Portsmouth diffusion plant which is now closed. The prototype lead cascade started operation in September 2007 and the test program with it to April 2010 refined the design of the AC100 centrifuge machines (which are much larger than the European Urenco centrifuges) to the "value-engineered AC100 machine", or ACR100 Mod 1, intended to be deployed in the commercial plant, and expected to deliver 350 SWU/yr per machine. An AC100 lead cascade started operation in March 2010 with plans to increase to 40-50 machines later in 2010. By the end of 2010, the company had spent $1.95 billion and required "additional capital beyond the $2 billion in Department of Energy loan guarantee program funding that it has applied for and USEC's internally generated cash flow." It said that another $2.8 billion was required from "financial closing of a loan guarantee", despite the benefit of using existing infrastructure.

The full plant was expected to commence commercial operation in 2010, and achieve full 3.8 million SWU annual capacity at the end of 2012. However, by July 2009 the project was suspended due to the DOE refusing to award a $2 billion loan guarantee, and asking USEC to withdraw its application. USEC refused to do this, and in July 2010, it resubmitted its loan guarantee application to the DOE, pointing out that $200 million commitment by Toshiba and Babcock & Wilcox in May 2010 to support the ACP also strengthened its

credentials[3]. Some of this was paid, but a stand-off then ensued pending loan guarantee, which was never issued.

The schedule in February 2011 was: begin commercial operations in May 2014; reach 1 million SWU/yr capacity in August 2015; and achieve annual capacity of about 3.5 million SWU in September 2017. In March 2012 USEC said its immediate RD&D program, dependent on government funding after May, was to “support building, installing and operating a 120-machine cascade and related support systems that will be replicated in 96 identical cascades in a full commercial plant.” This demonstration cascade was completed in April 2013 and became operational in October 2013 as “the centerpiece of the RD&D program with DOE”. By December it had completed 20 machine-years of operation of 340 SWU. USEC then obtained funds from the DOE under a cooperative RD&D program 80% funded by the DOE for this purpose, and extended to September 2014.

In November 2013 USEC said: "Despite the technical progress being made by the RD&D program, if funding is not in place at the end of the RD&D program or if USEC determines there is no longer a viable path to commercialization of ACP, we could make a decision to demobilize or terminate the project in the near term”. USEC said that its “prospects for adequate liquidity in 2014 are uncertain.”

With USEC (now Centrus) unable to proceed with the project, the intellectual property and the demonstration cascade reverted to the DOE under the terms of the cooperative agreement. In April 2014 the DOE said that in the light of commercial deployment of ACP being not viable at present, its Oak Ridge National Laboratory (ORNL) in Tennessee was taking over management of the project. The American Centrifuge Technology Demonstration and Operations Agreement between Centrus and ORNL ran to September 2015, backed by $97.2 million federal funding. It was then extended again for 12 months, with 60% reduced funding. Centrus said that this funding “excludes continued operations of America’s only operating cascade of advanced uranium enrichment centrifuges in Piketon,” which will therefore close down "after an investment of more than $2.6 billion in the project". In February 2016 Centrus announced that it had “completed operations of its demonstration cascade” of ACP at Piketon but would maintain its NRC licence. In March 2016 Centrus signed the US Centrifuge Technology Advancement contract with UT-Battelle, LLC, as operator of ORNL. The contract ran for six months. In September 2016 it signed a further contract with UT-Battelle "for maintaining

and advancing gas centrifuge uranium enrichment technology" over 12 months.

Earlier in March 2014 USEC applied for Chapter 11 reorganisational bankruptcy, a measure to enable it to restructure its debt while it continued operations. In its petition to the court, USEC listed assets of about $70 million and liabilities of $1.07 billion. The company emerged from bankruptcy at the end of September 2014 with the new name of Centrus Energy. About half of its workforce of 1300 have been working on the ACP project. The DOE notified Centrus in September 2015 that funding would be reduced below the level needed to keep the cascade operating. In mid-2016 Centrus commenced decontamination and decommissioning of the Piketon ACP plant, and expected this to continue through 2017.

Oak Ridge is where the ACP was originally developed to the 1980s. The DOE and NNSA said: “The government’s interest is in preserving and advancing the AC100 technology for possible future deployment for a national security mission.” The timeline for this is 2038 – see later subsection on Defence enrichment needs. The NNSA is effectively now the keeper of US enrichment technology. The domestic uranium enrichment capability is required because the USA can only use uranium for national security (defence) purposes that has been enriched by US-origin technology, hence excluding Urenco and SILEX. It is not clear that the USA needs any more HEU, but a related question is replenishment of tritium in nuclear warheads. This has been sourced from TVA’s Watts Bar 1 since 2003.

In June 2019 the DOE awarded a $115 million contract to American Centrifuge Operating (ACO), a subsidiary of Centrus Energy, to demonstrate the production of high-assay low-enriched uranium (HALEU), above 5% enrichment of U-235. The project involves construction of a cascade of 16 of the very large AC-100M centrifuges in Piketon, Ohio, where these machines were developed, to produce some 19.75% enriched uranium by June 2022. It should also demonstrate the capability to produce HALEU with existing US-origin enrichment technology, providing the DOE with a "small quantity" of it – understood to be less than one tonne per year – for use in research and development "and other programmatic missions". A number of new small reactor designs require HALEU and their developers are anxious to see a supply chain for it established. A government website said that ACO was the only source capable of executing the contract activities to meet the requirements of the DOE's HALEU Demonstration Program. In June 2020 the NRC accepted an application from Centrus to enrich

at up to 20% at Piketon and said that it expected to take a year to complete the review.

In mid-2021 it was licensed to produce HALEU up to 20%. In November 2022, the DOE awarded Centrus $150 million of cost-shared funding to finish the cascade, complete final regulatory steps, begin operating the cascade, and produce up to 20 kg of HALEU by the end of 2023 – this was delivered in November 2023. It was then expected that production would continue in 2024 at a rate of 900 kg of HALEU per year. However in February 2024 it was announced that delays in the supply of 5B cylinders from the DOE would prevent the delivery of this amount of material in 2024.

Eagle Rock Enrichment Facility

In mid-2007, Areva announced that it proposed to build a 3.3 million SWU/yr $2 billion centrifuge plant in the USA to supply domestic enrichment services. It submitted a licence application to the Nuclear Regulatory Commission (NRC) for this Eagle Rock Enrichment Facility in December 2008 and the NRC issued the licence in October 2011. Construction was to commence in 2012 at Idaho Falls, near the Department of Energy's Idaho National Laboratory, and operation was envisaged early in 2014, ramping up to full capacity in 2018. Areva Enrichment Services LLC, the owner and operator, signed a procurement & construction contract with URS in February 2011. It would be similar to Areva's new French plant (Georges Besse II). In 2009, Areva notified a planned doubling in capacity to 6.6 million SWU/yr, with the first stage being 3.3 million SWU/yr. In May 2010, DOE granted it a $2 billion loan guarantee. However, in December 2011 Areva announced that it was putting the project on hold as it seeks an additional investor, and in May 2013 the projected timeline became indefinite. In March 2017 Areva Nuclear Materials asked the NRC to terminate the licence "since the facility will not be constructed." (Areva Nuclear Materials is now Orano USA.)

Global Laser Enrichment: Paducah & Wilmington

In 2006 Silex Systems in Australia and GE Energy received US government approval for development in the USA of the third-generation SILEX (Separation of Isotopes by Laser Excitation) uranium enrichment process using laser technology. This approval cleared the way for development and eventual full commercial production under a licence agreement with Silex. It provided for GE

(now GE Hitachi, GEH) to construct in the USA an engineering-scale test loop, then a pilot plant or lead cascade, and expanded to a full commercial plant. GE was funding the development and making a series of payments to Silex. GE then said: "Commercialisation of the SILEX enrichment technology is a crucial part of GE's long-term growth strategy for the nuclear business." SILEX was rebadged as Global Laser Enrichment (GLE) which became the name of the project company. In mid-2008 Cameco bought into the GLE project, paying $124 million for a 24% share, alongside GE (51%) and Hitachi (25%).

GLE is now operating the test loop at Global Nuclear Fuel's (GNF's) Wilmington, North Carolina fuel fabrication facility. (GNF is a partnership of GE, Toshiba, and Hitachi.) In April 2010, GLE announced the completion of the first phase of the test loop programme and that technology validation would continue as designs for the commercial facility evolved and economic feasibility was verified. Further tests and ongoing activities accumulated performance, operating reliability and lifetime data on the technology, to assist the engineering design programme for the first planned commercial production plant. In April 2016 GE and Hitachi notified their intention to exit GLE. In February 2019 Silex Systems and Cameco agreed to buy out the GEH 76% share in GLE for $20 million on a deferred payment basis, so that Cameco holds 49% of GLE and Silex 51%. Cameco has an option to purchase an additional 26% of GLE. The agreement calls for Silex and Cameco to pay $300,000 per month to complete construction of the prototype enrichment facility, known as the Wilmington Test Loop, which has been partially built by GEH. Final US government approvals were granted in January 2021. Silex said the restructuring would "provide an ideal path to market for GLE and for the continued commercialisation of the SILEX technology in the US."

Wilmington: In mid-2009, GEH submitted the last part of its licence application for a commercial GLE plant at Wilmington. In February 2012 NRC found that its programs for the physical protection of special nuclear material and classified matter, material control and accounting provided an adequate basis for both safety and safeguards of facility operations. A full licence to construct and operate a plant of up to 6 million SWU/yr at WIlmington was issued in September 2012. GLE will now decide in the light of commercial considerations on whether to proceed with a full-scale enrichment facility there. The project, licensed to enrich natural uranium up to 8% U-235, could be operational in about two years, ramping up to annual capacity of 6 million separative work units (SWU) in the first

stage, and double that eventually. However, attention has turned to Paducah, and in July 2014 Silex reported that GLE was slowing down its activities at Wilmington.

Paducah: Silex Systems reported in November 2012 that there was a proposal for GLE to build an additional laser enrichment pant at Paducah, to enrich high-assay DU tails (above 0.34% U-235) owned by the DOE to produce 2000 tU/yr at natural uranium level (0.7% U-235). There is about 150,000 tonnes of these at Paducah and Portsmouth (among a total of 550,000 t tails). In August 2013 GLE submitted a proposal to the DOE to establish a "$1 billion" laser enrichment plant at Paducah, and in November the DOE announced that it would proceed with contract negotiations to this end, for enrichment of high-assay tails there. In January 2014 GLE told the NRC that it expected to apply for a licence to build, own and operate the Paducah Laser Enrichment Facility (PLEF). Negotiations with the DOE continued into 2016, and in November an agreement was signed with the DOE for it to sell about 300,000 tonnes of high-assay tails, justifying construction by GLE of the plant in the early 2020s. GLE expected licensing to take 2-3 years. The PLEF would become a commercial uranium enrichment production facility under an NRC licence, producing about 100,000 tonnes of natural-grade uranium over 40 years or more. The DOE would dispose of the reduced-assay balance. The estimated size is 0.5 to 1.0 million SWU/yr, since purchases of depleted uranium may not exceed 2000 t/yr natural uranium equivalent. In June 2020 the 2016 GLE agreement with the DOE was extended, and Silex said that commercial operation at 2000 tonnes of natural-grade uranium as UF_6 per year was expected in the late 2020s.

The Paducah GLE plant would serve a different market to that proposed for Wilmington, and would become a new source of secondary supply. In the long term it would compete with new mines, rather than other enrichment plants.

At the end of 2013, the DOE had about 13,000 tonnes of natural uranium, which it was bartering at the rate of 2400 t/yr to pay for decommissioning and clean-up of its Portsmouth and Paducah facilities. The proposed Paducah Laser Enrichment facility would provide a continuing source of such material beyond 2018.

Russian proposal

The contract signed in March 2011 between Techsnabexport (Tenex) and USEC Inc. for the delivery of uranium enrichment services during 2013-2022 gave effect to a memorandum of

January 2010 on the establishment of a joint venture in the USA to build a uranium enrichment plant based on Russian centrifuge technology. Tenex and USEC were undertaking a feasibility study on this, as a possible alternative to USEC American Centrifuge project. The main contract was for the supply of 21 million SWU to USEC over 2013-2022, worth $2.8 billion. See above subsection on USEC and Russian SWU.

US enrichment capacity expansion plans

	Site	Status	Capacity (SWU/yr)	Start-up
Urenco	Eunice	Under construction	700,000	2025
Orano	Oak Ridge	Proposed	"several million SWU"	early 2030s

Defence enrichment needs

The National Nuclear Security Administration (NNSA) needs a reliable supply of unobligated low-enriched uranium for LEU at 19.75% enrichment for research reactors, as well as for defence purposes, notably tritium production and HEU for naval reactors. There is no source of this at present, but in August 2015 the NNSA reported that it had enough unobligated LEU to get to 2038. By then it needed a plant of about 400,000 SWU/yr and favoured reviving the ACP technology in some form to provide this. Hence it is providing $32 million per year R&D at Oak Ridge to this end, with Centrus as contractor.

Decommissioning of old enrichment plants

The three original diffusion uranium enrichment plants – at Oak Ridge, Tennessee; Paducah, Kentucky; and Portsmouth, Ohio – were built by the federal government in the 1940s and 1950s for defence purposes and operated for 42, 60 and 47 years respectively. From 1969 to 1992, the DOE and its predecessor agencies sold some of the plants' enrichment services commercially under contracts that required utility customers to pay for future decommissioning and decontamination. Fees were collected for this but not set aside, and in 1992 the Energy Policy Act assessed utilities an annual fee of $150 million for 15 years, pursuant to which some $2.6 billion was paid. The administration's 2014 budget proposed levying further tax of $2.4 billion over 10 years on nuclear utility customers, which would have been the third time a

fee for the same programme was assessed. In July 2013 the National Association of Regulatory Utility Commissioners contested this.

Decommissioning of the Oak Ridge site was completed in October 2020, and the site was transformed into an industrial park.

Deconversion

Deconversion of the depleted uranium hexafluoride (UF_6) that remains as a byproduct after enrichment has not so far been undertaken on a large scale in the USA, and in fact for legal reasons depleted uranium is, in the USA, sometimes considered as a waste. There is about 765,000 tonnes of depleted UF_6 in the USA.

Uranium Disposition Services LLC (UDS), a joint venture of Areva, EnergySolutions, and Burns & Roe, was awarded a $558 million contract by the DOE in 2002 to design and build deconversion plants at Portsmouth, Ohio and Paducah, Kentucky. The contract ran to August 2010. The plants use a process developed by Areva which it employs at Richland, Washington and Lingen, Germany. The 13,500 t/yr Portsmouth plant was completed in mid-2010 and the 18,000 t/yr Paducah one in 2011. Operational testing and reviews were undertaken on both by EnergySolutions as managing partner. At Portsmouth about 250,000 t depleted UF_6 is stored, with another 75,000 t shipped from Oak Ridge, and at Paducah some 440,000 t depleted UF_6 is stored. Aqueous HF was a commercial by-product.

Then Babcock & Wilcox Conversion Services won a five-year $428 million contract from the DOE in December 2010 for uranium deconversion operations at both Portsmouth and Paducah, using these UDS plants. B&W Conversion Services (later: BWXT Conversion Services) is a joint venture of B&W Technical Services Group and URS set up in 2011 to operate the two DOE deconversion plants. B&W is the majority partner. Deconversion proceeded at about 20,000 tU/yr to 2016.

In September 2016 Mid-America Conversion Services (MCS), a joint venture of Atkins, Fluor and Westinghouse was awarded a $318 million, five-year contract to take over operation of the UDS plants at Portsmouth and Paducah from 2017. It was among five bidders for the contract. Atkins had been involved in building the plant. The DOE approved MCS to operate the UDS plants in February 2017. Products will be depleted uranium oxide for reuse or disposal and aqueous hydrofluoric acid for industrial use.

In December 2009, International Isotopes (INIS) subsidiary Fluorine Products Inc applied for a licence to build and operate a 6,500 t/yr deconversion plant and fluorine extraction facility 35 km west of Hobbs, New Mexico, and 70 km by road from the Urenco USA enrichment plant at Eunice. In April 2010, INIS signed a five-year agreement to provide toll deconversion services for DU tails from Urenco, from 2014. In October 2012 the NRC issued a licence to build and operate the $125 million plant, but in August 2013 the project was put on hold.

INIS hoped to start production in 2014, subject to raising $75 million capital. Initial capacity is for about 300 type 48Y cylinders, containing 3600 tonnes of UF_6 per year, increasing to about 575 (6950 t) in 2016. However in mid-2014 the company said that the project remained on hold until "additional uranium enrichment capacity comes on line that can provide us with additional opportunities to contract for deconversion service of that material." Some 1300 to 2300 tonnes of anhydrous hydrofluoric acid (HF) with 450 tonnes of fluoride gas will be produced per year for sale by INIS, and the depleted uranium belonging to the enrichment companies will be stored as more stable U_3O_8. In-mid 2010, the first part of a $65 million loan guarantee application by INIS was approved by the US Department of Energy (DOE), and the company has been invited to submit the second part of its application[2]

UUSA's licence amendment in March 2015 to 10 million SWU/yr enrichment capacity provides the potential for 1000 cylinders of depleted UF_6 per year arising from that source alone. However, the following month INIS said that "while we remain optimistic, at the present time we cannot predict if or when the plant will be built."

Preceding this proposal an agreement was signed in 2005 between LES (UUSA) and Areva to make use of the latter's technology in deconverting LES' DU tails. Areva NC has operated a small deconversion plant in association with its Richland fuel fabrication plant in Washington state.

Uranium fuel fabrication

The USA has three main fuel fabrication facilities to convert enriched uranium oxide into solid pellets for fuel rods. They are operated by Framatome (formerly Areva), Westinghouse, and Global Nuclear Fuel.

Westinghouse has a 1500 t/yr fuel fabrication plant for PWR and BWR fuel at Columbia, South Carolina.

Global Nuclear Fuel-Americas, owned by GE, Hitachi and Toshiba, has a 1500 t/yr plant at Wilmington, North Carolina.

In 2009, Areva's 35-year-old Richland, Washington fuel fabrication plant was the first to receive a 40-year licence extension from the NRC. It has 1200 t/yr capacity. In 2010, Areva and Mitsubishi Nuclear Fuel announced a 50-50 joint venture – US Nuclear Fuel – to manufacture APWR fuel at the Richland plant. This has apparently not proceeded.

BWX Technologies' Nuclear Operations Group subsidiary Nuclear Fuel Services (NFS) manufactures fuel for the US Navy at Erwin, Tennessee. It also downblends HEU.

MOX fuel fabrication

CB&I-Areva MOX Services (30% Areva, now Orano) was constructing the Mixed Oxide (MOX) Fuel Fabrication Facility (MFFF) at the US Department of Energy's Savannah River Site in South Carolina as part of the NNSA's Plutonium Disposition Program (see section below). The plant is based on Orano's Melox plant in France, but is more complex. The plant is designed to dispose of at least 34 tonnes of weapons-grade plutonium and convert it into useable fuel at the rate of about 3.5 t/yr, turning it into over 150 MOX fuel assemblies.

Construction of the plant began in 2007 under an onerous and complex regulatory regime set up by the DOE, and it was due to start operation in October 2016 and produce the first MOX fuel assemblies in 2018, for both PWR and BWR reactors. The cost estimate then escalated from $4.9 billion to $7.7 billion with operation beginning late 2019. The NRC extended the construction licence to 2025 to allow for a lower rate of funding. The cost escalation was initially blamed on approval of estimates before design was complete, and led to the plant facing shutdown in 2014 due to funding cut-off. The Energy Secretary in April 2014 said that NRC concerns had caused major changes in the plant's design from the French original. Funding has been $420 million in FY 2015 for the fissile materials disposition program including this plant, then $345 million for FY 2016, followed by $340 million for FY 2017. In May 2017 Areva said that $5 billion was required to finish construction, making a total cost of $10 billion, with commissioning in under ten years. In May 2018 CB&I MOX Services was acquired by McDermott.

In early 2016 the DOE sought to terminate the project and "downblend" the plutonium instead, so that it could be disposed at the Waste Isolation Pilot Plant (WIPP) in New Mexico. But Areva reminded the DOE that, according to their Plutonium Management and Disposition Agreement and 2011 amending protocol, the USA and Russia must change their surplus plutonium at the isotopic level to render the nuclear material unattractive for weapons use. "Though considerations have been raised about pursuing other non-proliferation options, none of the other methods change the plutonium's isotopic structure, so would require renegotiations with the Russians," Areva said. Russia made it plain that the downblending option would amount to a violation of the 2011 protocol, and in October 2016 Russia unilaterally cancelled the agreement.

In May 2018 the Energy Secretary said that he was terminating the project and proceeding with downblending the plutonium at Savannah River for disposal at WIPP. NNSA formally terminated the contract with CB&I Areva MOX Services in October 2018, and NRC terminated the permit for the plant in February 2019. NNSA expects that the diluted material would be in about 113,000 drums of 200 litres each, shipped over 2026 to 2050. "The department's independent cost estimate concluded that the remaining dilute and dispose lifecycle cost is $19.9 billion. The department estimated the remaining lifecycle cost of the MOX fuel programme to be $49.4 billion." The DOE is also "exploring whether any of the plutonium currently in South Carolina can be moved elsewhere for programmatic uses."

Simultaneously the under-secretary of defence for acquisition and sustainment, and the DOE under-secretary and NNSA administrator, said: "An evolving and uncertain geopolitical landscape calls for the United States to recapitalise its defence plutonium capabilities." They propose the MFFF be converted to produce or recondition plutonium pits, the core of nuclear weapons, at the rate of 50 per year.

In February 2010 the NNSA signed an agreement with TVA to evaluate the use of this MOX in its Sequoyah and Browns Ferry power plants, moving up to possibly 40% core load. Duke Energy has used four mixed oxide test fuel assemblies incorporating this weapons-grade plutonium (fabricated in France) at its Catawba 1 nuclear power reactor. This was to prepare for possible use of MOX for 20-40% of the cores of the Catawba and McGuire reactors from about 2010, using the fuel fabricated at MFFF.

Recycling reactor-grade plutonium in reactor fuel has been prevented in the USA by the banning of used fuel reprocessing (see below). However, Global Nuclear Fuel - Americas, a joint venture of General Electric, Toshiba and Hitachi, was preparing for licensing and production of MOX fuel for use in commercial BWR reactors by 2026.

Military surplus and other government stocks

As noted above (USEC and Russian SWU) almost half of the uranium used in US nuclear power plants from 1994 to 2013 came from Russian weapons-grade military uranium, downblended in Russia. Under this programme, by the end of 2013, 500 tonnes of high-enriched uranium (HEU) had been downblended into 15,259 tonnes of low-enriched uranium (LEU, average 4.4% U-235) for reactor fuel, representing some million SWU supplied for about $8 billion (paid by electricity consumers). The programme recycled about 20,000 warheads. According to Tenex, its total revenue was $17 billion, including hard currency gains and the cost of natural uranium component as contra supply.

On the US side, 174 tonnes of military high-enriched uranium was initially declared to be surplus and available for civil power generation. Much or all of this has been downblended by Nuclear Fuel Services in Tennessee, and the first fuel fabricated from it has been shipped to Tennessee Valley Authority (TVA) power plants.

In 2005, the Department of Energy's (DOE's) National Nuclear Security Administration (NNSA) announced that it was committing 40 tonnes of off-specification HEU in the USA to its Blended Low Enriched Uranium (BLEU) programme, with the fuel produced going to TVA power plants. To February 2012 about 22.7 t HEU had produced 301 t LEU (4.95% U-235), using natural uranium for blending.

In June 2007, the NNSA awarded contracts to Wesdyne International and Nuclear Fuel Services (NFS) to downblend 17.4 tonnes of HEU from dismantled warheads to yield about 290 tonnes of LEU, 230 t of which will be part of the new American Assured Fuel Supply programme.[b] The programme comprises 230 t of LEU at power reactor grade, equivalent to six reloads for a 1000 MWe reactor, according to NNSA. This LEU is available for use in civilian reactors by nations in good standing with the International Atomic Energy Agency that have good nonproliferation credentials

and are not pursuing uranium enrichment and reprocessing technologies, and now also to US utilities. The fuel – worth some $500 million – would be sold at the current market price.

In June 2009, NNSA awarded a further contract ($209 million) to NFS and Wesdyne for 12.1 tonnes of HEU, to yield some 220 tonnes of LEU by 2012. This batch of LEU was to provide fuel supply assurance for utilities which participate in the DOE's mixed-oxide fuel programme utilising surplus plutonium from US weapons. To cover the cost of the project, Wesdyne would sell a small part of the LEU (about 60 t) on the market over a three- to four-year period. (The scheme is consistent with international concerns to limit the spread of enrichment technology to countries without well-established nuclear fuel cycles. Russia has agreed to join the initiative.)

In March 2008, the DOE announced a policy for dealing with uranium which was surplus to defence needs. The 2008 inventory, totaling nearly 72,000 tonnes of natural uranium equivalent, was diminished by 2013, when the DOE issued a uranium inventory management plan update:

	Tonnes U 2008	Tonnes U 2013		Remaining at end of 2015, tU	Natural U equivalent end of 2015, tU
Unallocated U from US HEU inventory	67.6	18.0	HEU/LEU	10.4	1966
Allocated U from US HEU inventory	47.7	11.4	HEU/LEU	0	0
LEU	0	47.6	LEU	0	0
US-origin natural U	5156	5234	Natural U as UF_6	5234	5234
Russian-origin natural U*	12,440	7705	Natural U as UF_6	809	809
Off-spec NU/LEU non UF_6	4,461 (incl DU)	221	Natural U / LEU	221	617
Off-spec LEU as UF_6	NA	1106	LEU as UF_6	0	0
Depleted U > 0.34% U-235 from historic DOE enrichment	73,500	114,000	DU	114,000	25,000-35,000
Total					33,600-43,600

** Natural uranium exchanged under the 1993 agreement whereby Russian blended-down uranium is supplied to US utilities – effectively Russian-origin stocks*

The DOE 2008 plan showed a total of 22,700 tonnes of its uranium entering global markets before the end of FY2017, but with no more than 10% of US annual requirements being delivered to the market in any one year – apart from an allocation for the first cores of newly built US reactors.[c] This 10% guideline was removed in the 2013 revision to the plan. Transactions over 2008 to 2013 included: 4300 tU to USEC in payment for the clean-up of the Portsmouth enrichment plant, downblending of much HEU, 9082 t DU as UF_6 to Energy Northwest, disposal of much unsaleable off-spec material, and an addition of 47.6 tonnes LEU to inventory in 2012 from enrichment of Russian natural U.

In 2014 DOE confirmed that it would release 2705 tonnes of natural uranium equivalent annually to 2021, the same amount as in 2013, equivalent to 14% of US demand. It is mostly held as UF_6. In 2016 it is releasing 2100 tU – 1600 tU as natural UF_6 and 500 tU as LEU from downblending by NNSA contractors, Wesdyne, which was ongoing. Much of this will be to pay for Paducah and Portsmouth clean-up.

The stock of natural UF_6 could provide 1927 tU/yr for 7.5 years. After that, re-enrich the 73,500 tU in tails (av 0.375%) from the US stock of 464,000 tU in DUF_6 (including 295,000 tU at Paducah). The natural U value produced (@ $50/lb) is $3.4 billion, and it needs 14.7 million SWU to process and reduce tails to 0.20%.

The 114,000 tU of high-assay tails is DOE's largest remaining uranium asset in 2016.

In April 2016 NNSA awarded a $241.5 million contract to BWX Technologies' subsidiary NFS to downblend the last 10.4 tonnes of surplus HEU by 2019 to LEU, this being 1966 tonnes of natural uranium equivalent. In September 2018 TVA awarded NFS a $505 million contract to downblend 20.2 tonnes of HEU over 2019 to 2025 for LEU reactor fuel for Watts Bar 1 (approx. 4.4% U-235). The fuel is intended for use in the NNSA's Production of Tritium in a Commercial Light Water Reactor (CLWR) programme, where tritium is produced in TVA reactors using tritium-producing burnable absorber rods. TVA has been producing tritium for NNSA at Watts Bar 1 since 2004.

By 2016 there were a number of small modular reactor designs coming forward with a variety of fuel needs, some of them requiring LEU at almost 20% enrichment. The US Nuclear Infrastructure

Council (NIC) called for some of the downblending of HEU to create a "domestic strategic reserve" of high-assay LEU (HALEU), about 19.75% U-235. This would provide a small stockpile of fuel as a resource which would otherwise be very difficult to obtain (since civil enrichment plants normally cannot go above 5%). Others mentioned off-spec naval fuel as being a source of this. A reserve of 20 tonnes was suggested.

The NIC said that the only supply of fuel for many advanced reactors under development would otherwise be foreign-enriched uranium. "Without a readily available domestic supply of higher enriched LEU in the USA, it will be extremely difficult to conduct research on advanced reactors, potentially driving American innovators overseas."

In June 2018 the US Senate approved a proposal for a pilot programme to recycle used naval fuel for use in advanced reactors. There are currently no US-based commercial facilities capable of producing high-assay LEU.

In a submission to the DOE in September 2016, Urenco USA called for any government uranium stocks released to be at 19.75% enrichment, not around 5%, so as to avoid further depressing the already weak market. Uranium Producers of America (UPA) also suggested that the DOE-NNSA should be downblending HEU to levels between 5 and 19.75% LEU for research and advanced reactor fuel, thus minimizing the impact of DOE LEU below 5% on the commercial market.

See also information page on Military Warheads as a Source of Nuclear Fuel.

Plutonium disposition

In addition to the HEU surplus, the US government has declared 61.5 tonnes of weapons-grade plutonium to be excess to the needs of the US defence program. Of this, the government agreed under the 2000 US-Russia Plutonium Management and Disposition Agreement to dispose of 34 tonnes by 2014, incorporating it (with depleted uranium) into mixed oxide (MOX) fuel. A 2010 protocol to the agreement confirmed MOX for light water reactors as the sole disposal option for the USA, while Russia was to dispose of the material as MOX in fast reactors.

Construction of a MOX fuel fabrication plant at the DOE Savannah River site in South Carolina was authorised by NRC early in 2005

and began in August 2007 when funding became available. The MOX Fuel Fabrication Facility (MFFF) is being built by CB&I Areva MOX Services under a $2.7 billion contract to the DOE's National Nuclear Security Administration (NNSA), which will own the plant. (Most MOX plants use fresh reactor-grade plutonium comprising about one-third non-fissile plutonium isotopes; this uses weapons plutonium with more than 90% fissile isotopes.) The project includes a facility for plutonium pits dismantlement and metallic plutonium conversion to oxide which is complex and has delayed progress. It also includes a waste solidification building. The MFFF is expected eventually to produce about 1,700 MOX fuel assemblies from the 34 tonnes of weapons-grade plutonium under the Russia-US agreement – or more, should the government decide to dispose of some or all of the balance of its 61.5 tonnes surplus plutonium in this way. NNSA has said that it was considering options for a further 14.4 tonnes of surplus military plutonium, most of which could become MOX but with about 2 tonnes going to geological disposal in New Mexico.

However, early in 2014 the project was excluded from the DOE budget request for NNSA due to cost escalation, so faced being shut down while 60% complete. The timeline for operation had already gone out to 2019 and the cost estimate had blown out to $7.7 billion. NNSA's strategy for plutonium disposition is under the terms of the 2000 bilateral agreement with Russia, whose 60 t/yr MFFF at Zheleznogorsk started up in 2014. Another MOX plant at Dimitrovgrad has been expanded to similar capacity. The USA will then be in default on its bilateral agreement, while Russia has fulfilled its side. However the US Energy Secretary said that the USA remained committed to the agreement, and would assess alternatives to the project. Meanwhile CB&I Areva MOX Services is continuing to build it. In December 2014 the House passed a bill authorising $341 million for the plant, and this was followed by $345 million for FY 2016 and $340 million for FY 2017.

In April 2014 NNSA issued a report comparing alternative options for plutonium disposal. It said that the total cost of building and operating the Savannah River MOX plant would be over $31 billion, while burning the MOX in fast reactors would cost $58 billion, and immobilising it in ceramic or glass and storing it at the Idaho National Laboratory would cost over $36 billion. Downblending it and storing it at WIPP in New Mexico would cost only $16 billion. In March 2016 NNSA announced a decision to dispose of six tonnes of surplus non-pit plutonium at WIPP, after preparation including dilution at Savannah River, where it is now stored. The 6t is part of

13.1 tonnes of surplus plutonium for which a disposal pathway had not been decided.

In October 2016 the Russian President unilaterally announced cancellation of the 2000 Russia-US agreement to reduce each country's weapons-grade plutonium by 34 tonnes, primarily due to which the Savannah River MOX plant is being built.* He cited NNSA inclination or requests to dispose of the US plutonium in other ways than MOX. A draft bill was introduced to Russia's parliament, with an explanatory note saying that there were both political and technical reasons to cancel the deal, even though it could be renewed under certain conditions related to relaxing US sanctions. The presidential decree said the main reason for cancellation is "the emergence of a threat to strategic stability as a result of US hostile actions against Russia." The Nuclear Energy Institute then called on Congress to provide money for completing the Savannah River plant, since "allowing Russia to forgo its international obligation has significant national security implications for the United States and its allies."

* At the opening ceremony for Russia's MOX plant at MCC Zheleznogorsk in September 2015, Rosatom had compared its achievement in building its MOX plant for military plutonium in four years for RUR 9.3 billion ($142 million), with the US MFFF costing $7.7 billion over eight years to then, it being still incomplete.

As well as Duke Energy using four test fuel assemblies at its Catawba 1 reactor, weapons-grade plutonium in MOX test assemblies has been burned at the Saxton prototype reactor in the mid-1960s, and some MOX was burned in other US plants before 1977.

See also information page on Military Warheads as a Source of Nuclear Fuel and re plutonium, under Non-proliferation in the US Nuclear Power Policy paper.

Using enrichment tails

The US government owns some 700,000 tonnes of depleted uranium tails from past enrichment. In 2005 a pilot program treated over 8000 tonnes of these.

In 2011 ConverDyn and Urenco USA formed a partnership to offer its services to DOE in utilizing more of these DU tails. The Competitive American Tails Upgrade Partnership (CATUP) will use the two companies' conversion and enrichment facilities in

conjunction with its own natural uranium inventories to provide low-enriched uranium to the market while reducing the DU stockpile. CATUP's natural uranium concentrates would be converted to UF6 at ConverDyn's plant and exchanged for depleted UF6 from the DOE stockpile. This would then be cleaned up by CATUP, enriched by Urenco and sold.

Reprocessing used fuel

In 1977, the US government called a halt to the reprocessing of used fuel from commercial reactors as part of its stance against nuclear proliferation. The country has some experience with reprocessing oxide fuels as part of its military program, and has also built three civil reprocessing plants. The first, a 300 t/y plant at West Valley, New York, was operated successfully from 1966-72. However, escalating regulation required plant modifications which were deemed uneconomic, and the plant was shut down. The second was a 300 t/y plant built at Morris, Illinois, incorporating new technology which, although proven on a pilot-scale, failed to work successfully in the production plant. It was declared inoperable in 1974. The third was a 1500 t/y plant at Barnwell, South Carolina was built but not commissioned due to the changed government policy. It is now demolished. In all, the USA has over 250 plant-years of reprocessing operational experience, the vast majority being at government-operated defence plants.

One of these is the H-Canyon plant[d] at the DOE Savannah River site in South Carolina. This facility is the last such US plant able to treat used HEU fuel and similar materials that is still operational. From 2011 it is treating used HEU fuel from the USA and overseas research reactors which have been converted to LEU or shut down under the Global Threat Reduction Initiative. In 2014 it started a program to reprocess aluminium-clad and aluminium alloy used fuel from research reactors such as Oak Ridge's High Flux Isotope Reactor (HFIR) and to downblend the uranium product for use in power reactors such as TVA's*. The uranium separation part of this came online in August 2016, and the downblending plant is expected to start in 2018. Actinides and fission products will be vitrified for disposal. The vitrification plant at Savannah River had produced 7200 tonnes of vitrified wastes over 20 years to 2016, in about 4000 stainless steel canisters.

* In 2016 the plant had about 1000 fuel bundles and 200 HFIR cores to treat. In 2018 it was starting to downblend 120 HFIR cores for use in TVA reactors, notably Browns Ferry.

Several shifts in energy policy beginning in 2002, however, increased the likelihood of a resumption in reprocessing. In June 2005, the report accompanying the $31 billion energy and water funding bill approved by the Senate Appropriations Committee emphasised the need for new nuclear energy technologies. The DOE's Advanced Fuel Cycle Initiative (AFCI) would receive $85 million to develop fuel cycle technologies for Generation IV reactors including reprocessing and using fast neutron reactors to destroy long-lived components of wastes. A major driver for reprocessing was the reduction in the volume of high-level wastes, possibly obviating the need for any expansion of the planned repository at Yucca Mountain. In July 2005, with passage of the Energy Policy Act of 2005, the "recovery of the energy value from spent commercial fuel" became an explicit objective of the AFCI.

The shift on reprocessing was given further impetus by the government and industry commitment to develop of advanced nuclear reactors. In late 2005, the American Nuclear Society issued a position paper stating that "the development and deployment of advanced nuclear reactors based on fast-neutron fission technology is important to the sustainability, reliability, and security of the world's long-term energy supply."[4] An initial $50 million for "integrated spent fuel recycling facilities" focused largely on fast reactors was committed by the US Congress in 2006. The US industry body, the Nuclear Energy Institute, has said that the US nuclear industry needs to plan for recycling used fuel to reduce the long-lived radioactivity arising from it so that in a relatively short time high-level wastes become no more toxic than the original uranium ore. This means recycling and burning all the long-lived actinides, which is most efficiently done in fast neutron reactors.

The development of new reprocessing technology became a central element in the government's 2006 proposal for a Global Nuclear Energy Partnership (GNEP) to reduce the risk of nuclear proliferation (see information page on International Framework for Nuclear Energy Cooperation). Under this proposal, the US and other developed countries would develop proliferation-resistant recycling technologies and provide nuclear fuel to developing countries that promised not to engage in enrichment and reprocessing activities. GNEP attracted criticism, but it brought increased attention to the possibilities of reprocessing. Nevertheless, financial support for GNEP disappeared and by 2009, under Barack Obama's Democratic administration, the DOE removed its GNEP website and did not refer to the program in its budget request for FY 2010.[5] In June 2009, the DOE cancelled the programmatic environmental impact statement for GNEP "because it is no longer pursuing

domestic commercial reprocessing, which was the primary focus of the prior Administration's domestic GNEP program."[6]

Despite the lack of US government support for GNEP as such, research activities under AFCI have continued. A DOE-funded demonstration project involving several versions of the UREX+ process for spent fuel was underway at the Argonne National Laboratory. The DOE was also exploring reprocessing technologies such as AREVA's COEX, which is based on processes already used in France, the UK, Russia and Japan, as well as a number of other technologies that would require the widespread use of fast neutron reactors. (See our information page on Processing of Used Nuclear Fuel).

Areva costed plans for a major recycling complex in the USA, including reprocessing plant and MOX fuel fabrication plant, at $25 billion. It would have annual input capacity of 2500 tonnes of used fuel, and was expected to take 12-15 years to licence and build. The reprocessing cost was expected to be less than the 0.1 ¢/kWh fee charged for the nuclear waste fund.

A possible site for an initial reprocessing plant is at Morris, Illinois, which is the only licensed away-from-reactor wet used fuel storage facility in USA. It is adjacent to the Dresden nuclear power plant and currently stores about 700 tonnes of used fuel. It was the site of GE's Midwest Fuel Recovery Plant, a small reprocessing plant built in the early 1970s but not operated.

Department of Energy recommendations in January 2013 following the report of the Blue Ribbon Commission a year earlier excluded reprocessing and recycling.

In 2013 the NRC staff noted that Areva, “EnergySolutions, GE Hitachi Nuclear Energy, and Westinghouse Electric Company … are developing reprocessing strategies as part of an integrated fuel management strategy that includes research, development, and demonstration of advanced fuel utilization and recycling technologies.” However, NRC was not proceeding with developing regulations for reprocessing and the chairman said it might take 20 years to do so.

Decommissioning reactors

Nearly 30 civil prototype and commercial reactors have been decommissioned in the USA, and in 2013 four more joined them. Twelve have been totally dismantled (*Decon* option[e]) so that the

site is released for unrestricted use, notably Fort St Vrain, Big Rock Point and Shoreham. Owners of single-unit plants generally opt for Decon. Ten are in various stages of dismantling, or *Safstor*[f]. The Safstor option is generally best for units on sites with continuing operation, allowing for the growth of funds.

The Nuclear Energy Institute reported in 2006 that of the total $32 billion estimated to decommission all eligible nuclear plants at an average cost of $300 million, about two-thirds had already been funded. The remainder would be funded over the next 20 years (US nuclear plants are licensed for 40 years initially, and more than three-quarters have had licences extended to 60 years)[f]. See also appended Table.

The NRC regularly reports on the adequacy of decommissioning funds set aside during operation into a trust fund or similar. In 2013 it confirmed that commercial reactor operators had sufficient funds for decommissioning their plants, and that the formula for estimating costs is accurate. The total amount accumulated by all licensees was $45.7 billion at the end of 2012. In April 2016 the California Public Utilities Commission approved $4.41 billion in decommissioning costs for San Onofre 2 and 3, from funds held in trust.

In October 2016 a report by S&P Global Ratings said that no US nuclear utilities were facing a "serious funding gap" in meeting future reactor decommissioning costs, even in the light of the growing number of early reactor shutdowns. "The NRC's minimalist approach results in healthy surpluses in most cases and no negative results for utilities," it said, and that NRC requirements ensure "greater confidence that funds will be available and that the risk of a significant shortfall is remote." It pointed out: "Utilities regularly update decommissioning cost studies and file to adjust rates based on those studies, so there's an ongoing process for replenishing funds during a plant's operating phase."

See also appended Table of shutdown US reactors.

Used fuel and nuclear waste

US policy since 1977 has been to forbid reprocessing of used fuel and to treat it all as high-level waste. By the end of 2016, 78,591 tonnes of used fuel had been discharged from US reactors, with annual additions of over 2000 tonnes expected.* About 40% of the used fuel is in dry storage at reactor sites, including sites with no operating reactor.

A Government Accountability Office report in November 2014 said that provisions in the Nuclear Waste Policy Act authorizing DOE to establish consolidated storage facilities have either expired or are unusable because they are tied to progress with the Yucca Mountain repository project, which the DOE aborted in 2010.

* This compares with the American Coal Ash Association's estimate of 115 million tonnes of waste resulting from coal-burning factories and generators each year.

US policy is enshrined in the Nuclear Waste Policy Act of 1982 which established federal responsibility for all civil used fuel, including a timetable and procedures for the building of two repositories, funded by fees from utilities. It obliged the federal government, through the DOE, to begin removing the used fuel from nuclear facilities by 1998 for disposal in a federal facility. The Act was amended in 1987 to designate Yucca Mountain in Nevada as the sole initial repository for 70,000 tonnes of high-level wastes[g].

Yucca Mountain and federal responsibility from 1987

Despite several delays to the program[h], in June 2008 a construction licence application was eventually submitted to the Nuclear Regulatory Commission (NRC) by the DOE. However, following the 2009 presidential elections, the Barack Obama administration attempted to abort the Yucca Mountain project[i]. A high-level 'Blue Ribbon' commission was then appointed to come up with alternative proposals by the end of 2011[j]. The NRC terminated licensing activities for Yucca Mountain in 2010-11, with the multi-volume safety evaluation report (SER) written and then undergoing review.*

* Yucca Mountain has famously been opposed by an influential Nevada senator, Harry Reid. However, Nevada has not always been opposed to the project. In 1975 the state legislature passed a resolution “strongly urging” DOE’s predecessor agency (the Energy Research and Development Administration) to choose the Nevada Test Site – where Yucca Mountain is located – for the disposal of nuclear wastes. In 2012 Nye County wrote to the secretary of energy agreeing to host the repository in line with the 'Blue Ribbon' commission recommendations for consent-based siting of such facilities. One of nine sites originally identified by the Department of Energy in 1983, Yucca Mountain was the highest ranked of five nominated for further consideration and, in 1986, was the best of three remaining sites, based on its geohydrology, geochemistry,

rock characteristics, tectonics, meteorology, costs and socioeconomic impacts.

Before budget cuts and policy announcements by the Obama administration, the DOE estimate of when Yucca Mountain repository might be operational was about 2021,[9] with some expansion of the original 70,000 tonne capacity[10]. The total cost in mid-2008 was put at about $96 billion (in 2007 dollars) for its construction, operation for 110 years, decommissioning from 2133 and the transport of used fuel to it[11].

In August 2013 the federal Appeals Court ordered the NRC to resume its review of DOE's application for a licence to construct and operate the Yucca Mountain repository. The appeal was brought by the National Association of Regulatory Utility Commissioners, two states and others who argued that NRC ignored its statutory responsibility when it terminated its review of DOE's 2008 application. The court said the case "raises significant questions about the scope of the Executive's authority to disregard federal statutes. The case arises out of a longstanding dispute about nuclear waste storage at Yucca Mountain in Nevada. The underlying policy debate is not our concern. The policy is for Congress and the President to establish as they see fit in enacting statues," and for the president and executive agencies to implement, it said. The court said that its task is to ensure that agencies such as NRC comply with the law.

The Nuclear Energy Institute and the Nuclear Waste Strategy Coalition said that this ruling on the used fuel repository licensing case was a clear signal that the NRC was obligated to complete its evaluation of the DOE application and issue a final decision granting or denying the licence. The August 2013 order for the NRC to restart the licencing involved the NRC finishing its review of the safety evaluation report (SER) for the repository. The NRC complied and requested the DOE to prepare a supplementary environmental impact statement (EIS) for it. Volume 3 of the SER, a technical evaluation, was published in October 2014 and confirmed that Yucca Mountain would meet all regulatory requirements for a repository. Volumes 2, 4 and 5 of the SER were published by January 2015, completing the NRC's technical safety review of the DOE licence application for the repository and confirming that its requirements were met. The DOE published the NRC's draft supplementary EIS in November 2015 confirming the suitability of the site, and in May 2016 the NRC issued its final supplement to the EIS, analysing potential impacts on groundwater and surface groundwater discharges and determining that all impacts would be negligible.

The DOE's Office of Civilian Radioactive Waste Management that oversaw the Yucca Mountain program was closed in 2010 when the administration withdrew the licence application. Following NRC's safety evaluation review, the DOE now needs to revive its Yucca Mountain activities. While provisions in the 1982 Nuclear Waste Policy Act authorized the DOE to establish consolidated storage facilities, a Government Accountability Office (GAO) report in November 2014 said that these provisions had either expired or were unusable because they were tied to progress with the Yucca Mountain repository project, which the DOE aborted in 2010. In May 2017 a GAO report said that the DOE, NRC and non-federal parties will need to rebuild their organisational capabilities, including the rehiring or recruiting of legal, scientific and other experts. There were also several key legal steps required.

Storage at nuclear plants, waste fees and damages

With repository development derailed, storage space at some operating nuclear reactors ran out, and at most of the 65 nuclear power plant sites (60 operating, in 2017) pool storage is being supplemented with dry cask storage. Of the total inventory of 78,600 tonnes of used fuel at 74 reactor sites in 35 states[k], about 40% was in dry cask storage at the end of 2016. The total increases by 2000 to 2400 tonnes each year. In July 2015, 23,000 tonnes of used fuel was in dry storage at 67 sites, using 2159 dry storage casks (2463 casks at end of 2016). As of the end of 2019 the total inventory of used fuel was 84,000 tonnes.

Holtec-HI-STORM-casks-at-Vermont-Yankee-(Holtec).jpg

Used fuel casks at Vermont Yankee (Holtec)

A number of utilities have sued the federal government for not meeting its obligation under the 1982 Nuclear Waste Policy Act to begin taking waste by 1998 and have been awarded damages by the courts[l]. The DOE reported that a total of $6.1 billion had been paid from the taxpayer-funded Judgment Fund to 30 September 2016. Government liability for breach of contract is estimated by the DOE to be $30.8 billion, including the amounts paid, assuming the DOE starts to accept used fuel in 2021. The Government Accountability Office reported that almost $9 billion had been paid from the taxpayer-funded Judgment Fund to 2020. The remaining government liability for breach of contract and interim storage was estimated by the DOE to be $30.6 billion in 2020.

By the end of 2016, utilities had contributed over $21.2 billion into the Nuclear Waste Fund (NWF), which attracts interest at about $1 billion per year. From it there have been some $7 billion as funding disbursements for the Yucca Mountain program, leaving about $28 billion.[m] The fund received over $750 million in fees each year. In November 2013 a federal appeals court unanimously ruled that the DOE should cease collecting the fees from utilities. "Because the Secretary [of Energy] is apparently unable to conduct a legally adequate fee assessment, the Secretary is ordered to submit to Congress a proposal to change the fee to zero until such a time as either the Secretary chooses to comply with the [Nuclear Waste Policy] Act as it is currently written, or until Congress enacts an alternative waste management plan." The Nuclear Energy Institute said that "the court's decision should prompt Congress to reform the government's nuclear waste disposal program. We strongly encourage Congress to establish a new waste management entity, and endow it with the powers and funding necessary to achieve the goals originally established in the Nuclear Waste Policy Act" of 1982. The DOE stopped collecting the waste fees in May 2014. The Nuclear Energy Institute estimates that the NWF has about $35 billion in 2016.

Managing the stand-off, new approaches

Under new standard contracts with DOE, proponents of new reactor construction must undertake to store used fuel on site indefinitely, so that the DOE does not become liable for delays. The contracts specify that the DOE will begin removing used fuel within 20 years of the first refueling. As of January 2009, 19 such contracts had been signed under the Nuclear Regulatory Commission's (NRC's) **Waste Confidence Rule**. They were a prerequisite for new reactor licensing and for licence renewals, and reflected the degree to which the NRC is confident that used fuel from US power reactors could be safely managed. The rule stated that the NRC is confident that a repository (not necessarily Yucca Mountain) will be available 50 to 60 years after a reactor operating licence expires. A court ruling in June 2012 set back the process and required further analysis, resulting in revised documentation in July 2013. The revised **continued spent storage rule** with supporting generic environmental impact statement (GEIS) was adopted by NRC in August 2014, and came into effect in October 2014.

The so-called Blue Ribbon Commission's interim report to Congress in January 2012 recommended the development of centralized interim storage, establishing a new organization outside DOE to

manage the US used fuel program, assured access to the nuclear waste fund in a trust rather than treasury, continued pursuit of a geological repository, and continued debate on reprocessing used fuel for recycle (strongly supported by industry). More generally the report said that "this nation's failure to come to grips with the nuclear waste issue has already proved damaging and costly. It will be even more damaging and more costly the longer it continues." Congress expected the DOE to develop a new strategy for managing used nuclear fuel and other nuclear waste within six months in response to the report. The US industry welcomed the report.

In January 2013 the DOE announced a new approach based on the report, including setting up a new organisation to manage the siting, development and operation of the future waste stores. It envisaged a consent-based approach for setting up a consolidated interim storage facility (CISF), with a 'pilot interim store' being in operation in 2021, with a priority on taking used nuclear fuel from current shutdown power plant sites. By 2025 a larger 'full-scale interim store' would open, and by 2048 an underground disposal facility should be in place to permanently store and dispose of the material. The mandate for the new organisation would exclude reprocessing of used fuel. However, there has been no progress regarding this organisation. Under the new administration, in 2017 the draft consent-based approach for siting interim storage was abandoned in favour of reviving the Yucca Mountain project.

In October 2014 a DOE report recommended development of a separate repository for military HLW, rather than a joint facility for civil and defence wastes. Nuclear industry spokesmen opposed it, saying that it would further delay moving spent fuel from reactor sites. However, in March 2015 the President authorized the DOE to plan a separate repository for military high-level radioactive waste, and the DOE said it would concurrently undertake a phased, adaptive consent-based approach to siting a large interim storage facility for civil used fuel. (See further below.)

An October 2014 DOE report said: “Preliminary evaluations of deep borehole disposal indicate a high potential for robust isolation of the waste, and the concept could offer a pathway for earlier disposal of some wastes than might be possible in a mined repository.” It was not considering this option for the main volume of civil used fuel.

In May 2017 the administration’s budget proposal for FY 2018 included $120 million to restart NRC licensing activities at Yucca Mountain, and $30 million was allocated from the Nuclear Waste

Fund. The NRC expects to take 3 to 5 years before it can authorize construction.

In May 2018 the House of Representatives voted decisively (340 to 72) in favour of reversing former president Obama's veto of 1982-legislated plans for disposal of the US inventory of used nuclear fuel and other high-level waste. The Nuclear Waste Policy Amendments Act 2018 also provides for setting up privately-owned central interim storage facilities meanwhile, increases the planned repository size from 70,000 to 110,000 tonnes and rules out a separate repository for military HLW. A Nevada amendment designed to weaken the bill failed (80 to 332).

However, in September 2018 a funding package agreed by a Senate and House conference committee for financial year 2019 excluded any provision for restarting the licencing of the Yucca Mountain repository. The bill passed the Senate in a 92-5 vote, cleared the House 377-20 the following day and was signed into effect despite expressed White House disappointment regarding the exclusion of Yucca Mountain funding.

In September 2021 a US Government Accountability Office (GAO) report said that Congress needed to take action to break the impasse over a permanent solution for commercial used nuclear fuel. It suggested further measures to develop a consent-based process for both consolidated interim storage and final disposal of used fuel.

Other initiatives for used fuel

As well as the DOE Yucca Mountain enterprise, Private Fuel Storage LLC (PFS) planned to store used fuel on a site in Utah for up to 40 years pending disposal. PFS is a consortium of eight utilities impatient with the DOE and applied for a repository licence in 1997. In February 2006, the NRC issued a 20-year licence for a 40,000 tonne centralised surface dry storage facility on land owned by the Skull Valley band of the Goshute Indians[n]. But ongoing state government opposition led to the Department of Interior then disapproving the Goshute-PFS lease and the use of public land as a transport corridor to the planned facility. This decision was appealed in 2007, and in July 2010 the "arbitrary and capricious" 2006 Department of Interior ruling was overturned by a District Court ruling (which was not challenged). However the Department's Bureau of Indian Affairs maintained its opposition and in December 2012 PFS sought the termination of the 2006 licence due to the cost of maintaining it and the fact that all the PFS

members now have dry storage at their plant sites. Construction time would have been 24 to 36 months, using Holtec's Hi-Storm ventilated dry storage system. In 2015 PFS commenced moves to dissolve the entity.

In the light of the 2006 Utah setback, the Nuclear Energy Institute started a search for communities willing to host interim storage sites for used fuel. It received several offers and by mid-2008 had reduced the possibilities to two, and discussions were proceeding. A commercially-operated facility on a 400 ha site was envisaged for each location, with the DOE paying rent for casks stored there. The concept was picked up in a bipartisan 2013 Senate bill but did not proceed.

In response to the January 2013 DOE strategy calling for a consent-based approach for setting up a consolidated interim storage facility (CISF), two proposals have come forward, in New Mexico and Texas. The NRC has proposed a 42-month review period for each application, after which the applicants will be authorized to construct and operate those facilities. The applicants will separately need to negotiate contractual terms with the DOE on transferring title of the used fuel, transportation and other issues.

Orano NUHOMS for Waste Control Specialists, TX

The first consolidated interim storage facility (CISF) proposal was from some Texas counties which expressed interest. In January 2015 local government unanimously approved a plan by Waste Control Specialists (WCS), which operates the Texas Compact facility (see below), to develop a CISF for used nuclear fuel at the low-level radioactive waste disposal facility the company has operated in Andrews county since 2012. The resolution references a March 2014 analysis by the Texas Commission on Environmental Quality that concluded consolidated storage of used fuel and high-level radioactive waste in Texas would result in "considerable savings" to electricity consumers compared with storage at each nuclear power plant. WCS applied to the NRC in April 2016 to build and operate an above-ground used fuel dry storage container facility at the 60 sq km Andrews County site, on the New Mexico border and very close to the Urenco USA enrichment plant at Eunice, south of Hobbs. This will be initially for used nuclear fuel from Texas' nuclear facilities, and decommissioned reactors across the USA, and was expected to operate from 2020.

In May 2015 WCS signed an agreement with TN Americas (storage and transport division of Areva Nuclear Materials, now Orano USA) for the design, development, construction, operation and maintenance of the Texas CISF. Areva would also support the transport of used fuel to it. NAC International then joined Areva in supporting the proposal and seeking a licence for it. The CISF would provide above-the-surface storage for used fuel which has already been placed in dry storage casks at the nuclear plant site. The NUHOMS storage casks from Areva TN can accommodate high-burnup fuel. They would be encased in an additional NRC-certified transportation cask and transferred primarily by rail to the facility, where they would be removed from the transportation cask and placed in storage, in horizontal vaults – the NUHOMS horizontal storage module. The CISF would have 40,000 tonne capacity, developed in 5000-tonne increments. The proposal includes opportunities for 20-year licence renewals after the initial 40-year licence period.

However, in April 2017 WCS asked the NRC to suspend its review of the application pending its purchase by EnergySolutions, if this was approved (see *Low-level waste section* below). Its then owner, Valhi Inc, was not supportive and wrote off costs incurred on the project. In June 2017 a US court barred the WCS sale on antitrust grounds. WCS said that the NRC's $7.5 million cost estimate for reviewing the application was much higher than expected. In January 2018 WCS was acquired by an investment affiliate of private equity firm JF Lehman & Company. In March 2018 WCS and Orano USA announced a joint venture – Interim Storage Partners (ISP) – to complete licencing the project. In May 2020 the NRC issued a draft environmental impact statement for consolidated interim storage of up to 5000 tonnes of spent fuel for 40 years and a final EIS in July 2021. In September 2021 it issued an operating licence "to receive, possess, transfer and store up to 5000 tonnes of spent fuel". Seven expansion phases over 20 years are envisaged to take capacity to 40,000 tonnes. In November 2020 the state governor had urged the NRC to deny the licence.

Orano's TN Americas and NAC between them represent 62% of existing dry storage systems in the USA, including 78% of the used nuclear fuel stored at sites where there is no longer an operating reactor. The licence request references dry storage casks designed and certified by Orano USA subsidiary Transnuclear (Orano TN) and by NAC International. ISP references over 460 NAC casks and 1265 Orano TN casks storing spent fuel at both operating and decommissioned US nuclear plants as of June 2019.

Holtec Hi-Store CISF for Eddy-Lea Energy Alliance, NM

The second consolidated interim storage facility (CISF) proposal was from the Eddy-Lea Energy Alliance (ELEA), a consortium of Eddy and Lea counties and Carlsbad and Hobbs cities, which came forward with a proposal supported initially by the state governor. ELEA has a 4 sq km site between Carlsbad and Hobbs in the southeast of New Mexico, about 15 km north of the Waste Isolation Pilot Plant (WIPP). Holtec has signed an agreement with the Eddy-Lea Energy Alliance LLC to establish a Hi-Store CISF there, requiring less than 25 hectares. The project includes the design, licensing, construction and operation of an interim storage facility using an enlarged version of Holtec's Hi-Storm UMAX storage system, already deployed at two US nuclear power plant sites. This stores used fuel in sealed canisters inside ventilated vertical steel and concrete containers 5 m high and below ground, with 11-tonne lids. The containers are set up in a 7.7 m deep excavation and low-strength concrete grout is backfilled around them. Canisters can be retrieved in a few hours at any time. (Holtec said that only 13 ha would be required to store 75,000 tonnes of used fuel, as envisaged for the suspended Yucca Mountain repository.)

In March 2017 Holtec applied to the NRC for a licence for phase 1 of the Hi-Store CISF, which would eventually cover 120 ha and hold 10,000 canisters of spent fuel for 40 years. Phase 1 would be for 500 canisters, with 8680 tonnes of spent fuel and was hoped to be operational in 2020. Nineteen subsequent phases would be for the rest of the capacity. The CISF would use Holtec's already-certified Hi-Storm UMAX system "designed to accept every canister currently loaded at every US nuclear power plant." Holtec in August 2016 had applied for NRC licensing for using Orano's NUHOMS canisters in the Hi-Storm UMAX facility. The NRC expects licensing to take until May 2021.

Military HLW and WIPP

The USA also has some military 'High-Level Wastes' (HLW) which will need disposal with the civil high-level wastes such as used fuel. In fact these are all or mostly fairly old, and the radioactivity has decayed to levels comparable with those of the transuranic (TRU) wastes being disposed of at the Waste Isolation Pilot Plant (WIPP) in the Salado salt formation in New Mexico since 1999, near Carlsbad. By 2013 WIPP held about 85,000 m^3 of TRU wastes. Its original projected closure date of 2030 is likely to be extended to

2055, especially following its closure due to a leak of radioactivity in mid-February 2014*. WIPP was reopened in January 2017, received its first new shipment of waste in April – from Idaho – and mining operations are expected to resume before the end of the year. Recovery and modifications cost more than $500 million, according to the DOE recovery plan. About $242 million would be required to bring WIPP back into initial operations, plus $77 million to $309 million for infrastructure upgrades, including an improved ventilation system, to make it fully operational again.

* One drum in storage ruptured due to an exothermic chemical reaction in organic absorbent material. See April 2015 Accident Investigation Report.

In March 2015 the president authorized the DOE to move forward with planning for a separate repository for high-level radioactive waste resulting from atomic energy defence activities. The DOE will undertake a consent-based approach to siting storage and disposal facilities, as called for in the administration's 2013 *Strategy for the Management and Disposal of Used Nuclear Fuel and High-Level Radioactive Waste*, and building upon the work of the bipartisan Blue Ribbon Commission. Most defence waste has decayed to the extent that it could be placed at WIPP. Treating it separately from civil HLW is expected to assist progress with the latter. DOE Factsheet.

The USA's only high-level radioactive waste vitrification plant has been operating at the Department of Energy's Savannah River site since 1996. The Defense Waste Processing Facility converts waste-containing sludges into borosilicate glass, which is currently stored onsite in stainless steel canisters. By 2016 the facility had processed over 4000 canisters of vitrified waste since operations began, and it is expected to operate for a further 20 years.

Low-level waste

For low-level waste (LLW)º, there are a number of specialist US facilities which are licensed and regulated by the host states under terms of agreements with NRC. Otherwise LLW storage is at reactor sites:

1. Waste Control Specialists operates the Texas Compact Facility, which has taken some LLW since 1998 and was licensed in 2009 for class A, B & C LLW, and also to take Vermont's and DOE LLW. The company was licensed in 2012 to take LLW from other states, and in October 2011 it entered into an agreement with Utilities Service Alliance, representing 15 utilities

including 17 nuclear power plants with 27 reactors. A separately licensed site in the complex handles radioactive waste from federal sites around the country, and in 2013 started receiving shipments from Los Alamos Laboratory in New Mexico. The two facilities in Andrews County are 580 km west of Dallas, near the New Mexico border, and owned by the state.
The Texas Compact Facility is the only one in the USA to be licensed within the last 30 years that is authorised to dispose of all classes of low-level radioactive waste materials considered suitable for near-surface burial. This includes depleted uranium, classified as class A LLW. Since the radioactivity of this will increase over time, it will be encased in concrete and buried 30m deep. It accepts waste from the nuclear medicine, research, and power sectors. Some $500 million was invested in setting up the facility. In 2014 it gained approval to expand capacity from 65,000 m^3 to 255,000 m^3.

2. EnergySolutions' facility at Clive, Utah, which accepts class A LLW (about 90% of all LLW) from all over the USA (a proposal to take low-level decommissioning wastes from Italy was abandoned in mid-2010). It has applied for a licence to accept depleted uranium*, which is considered class A LLW by NRC, and envisages burying up to 700,000 tonnes of it about 3 metres deep.
3. EnergySolutions at Barnwell, South Carolina, for Class A, B & C LLW from that state, New Jersey and Connecticut. It has operated since 1971, and also processes high activity filters from the nuclear industry.
4. EnergySolutions at Bear Creek, near Oak Ridge, Tennessee for safe processing and packaging of radioactive material for permanent disposal. It claims to be the largest licensed commercial US LLW facility with innovative technologies for radioactive material volume reduction (compaction, melting, incineration). There is nearby capacity for recycling depleted uranium.
5. American Ecology Corp at Richland, Washington accepts Class A, B & C waste from the Northwest and Rocky Mountain compacts.

* It would be disposed with other LLW so that its natural increase in radioactivity over thousands of years is matched by the decrease in that of the other LLW, so that the disposal cells remain Class A forever.

Notes & references

Notes

a. In August 2010, the Department of Energy awarded a $2.1 billion contract to a joint venture between Fluor Corp and Babcock and Wilcox (B&W, now BWXT) for decontamination and decommissioning of the huge (15 sq km) Portsmouth Gaseous Diffusion Plant (GDP) uranium enrichment site in Ohio, from March

2011. In March 2016 the contract was extended by 30 months. From the 1960s to 2001 the Portsmouth plant worked in tandem with its sister facility in Paducah to enrich uranium for use in power plants. The Paducah plant enriched uranium up to 2.75% U-235 and then shipped it to Portsmouth for further enrichment to some 4-5%. [Back]

b. NFS diluted the material in Tennessee to yield some 290 tonnes of low-enriched uranium (4.95% U-235) by early 2012. Wesdyne, the prime contractor, took 60 t of this LEU as payment in kind, then stored the rest at the Westinghouse fuel fabrication plant in South Carolina to be available for the Reliable Fuel Supply programme – an international fuel reserve – which then became the American Assured Fuel Supply (AFS) programme. The DOE has said it would maintain a uranium reserve of 670 tonnes U – equivalent to about 20 power reactor reloads – for energy security reasons. This will be kept as low-enriched uranium stored either at the DOE's Portsmouth or Paducah sites, or might be kept as part of a commercial entity's working inventory. This intention may have been subsumed under the AFS programme. [Back]

c. In line with this, the annual quantity coming from dismantled weapons and re-enriched depleted uranium increases steadily to reach 1920 tonnes U in 2013 and then continues at that level, totalling 15,000 tonnes U. From 2010 to 2015, another 7700 tonnes U from Russian-origin stocks is allocated for the first cores of newly-built reactors in the USA. [Back]

d. H-Canyon dates from 1955 and originally recovered uranium, neptunium and plutonium from used military and research reactor HEU fuel. It also recovered neptunium-237 and plutonium-238 from special irradiated targets, and played a vital role up to 2008 in the production of the Pu-238 used to power numerous deep space exploration programs. Since 1998 it has recovered HEU from degraded materials and spent fuel, to recycle it as LEU as part of the DOE's environmental management program. This program will continue to 2024. [Back]

e. In the USA, the option of immediate decommissioning of nuclear plants is known as the Decon strategy. The second option, where a facility is dismantled after allowing much of the radioactivity to decay, is known as Safstor. The third option, known as Entomb, is to permanently encase radioactive contaminants onsite until the radioactivity has decayed to a level where restricted release of the facility is possible.

US plants with Decon completed are: Big Rock Point, Elk River, Fort St Vrain, Haddam Neck, Maine Yankee, Pathfinder, Rancho Seco, Saxton, Shippingport, Shoreham, Trojan and Yankee Rowe. In 2011 Decon is in progress at Fermi 1, Humboldt Bay 3 and San Onofre 1. Those plants in Safstor include Dresden 1, Indian Point 1, LaCrosse, Millstone 1, Peach Bottom 1, and Zion 1&2, as well as *NS Savannah*. Three Mile Island 2 is in post-defuelling monitored storage. The only US plants subject to the Entomb option are three small experimental ones.

See the Fact Sheet on Decommissioning Nuclear Power Plants on the Nuclear Regulatory Commission's website (www.nrc.gov) for more information. [Back]

f. For further information, see the page on Sites Undergoing Decommissioning on the Nuclear Regulatory Commission's website (www.nrc.gov); see also the information page on Decommissioning Nuclear Facilities, which has some details of US plants. [Back]

g. The 70,000 tonne high-level waste repository planned at Yucca Mountain would take 63,000 tonnes of used reactor fuel, 2333 t of naval and Department of Energy used fuel and 4667 t of other high-level wastes, all from 126 sites in 39 US states. Studies by the Electric Power Research Institute show that the repository could hold at least 260,000 tonnes and possibly 570,000 tonnes of used fuel and high-level wastes, rather than the arbitrary 70,000 tonnes set by Congress in 1982.[7] [Back]

h. After several years of failure to get matching bills through both houses of Congress, early in 2000 the House of Representatives finally passed the Nuclear Waste Policy Amendments Act 2000 by 253 votes to 167, matching the earlier Senate passage of the legislation by 64 to 34. However, the President then vetoed it. The Bush Administration sought to make some urgent headway on the matter, and several reports in 2001 suggested no insurmountable scientific or technical problems with the proposed repository site in Nevada. The US Energy Secretary recommended that the site be approved as the nation's permanent repository. This was strongly supported by Congress and signed into law in July 2002. The Department of Energy submitted a licence application to the Nuclear Regulatory Commission in June 2008. Less than a year later, the new Obama administration's FY 2010 Congressional Budget Request confirmed "the Administration's decision to terminate the Yucca Mountain program while developing nuclear waste disposal alternatives." [Back]

i. The first budget from the Obama administration in 2009 cut off most of the money for the Yucca Mountain project[8]. Although the Administration said it had cancelled the project, the Atomic Safety and Licensing Board of the Nuclear Regulatory Commission ruled in June 2010 that the construction licence application submitted two years earlier may not be withdrawn without consent of Congress. See the Nuclear wastes section in the information page on *US Nuclear Power Policy* for further information. [Back]

j. In January 2010, the DOE announced the formation of a 15-member *Blue Ribbon Commission on America's Nuclear Future* to "provide recommendations for developing a safe, long-term solution to managing the nation's used nuclear fuel and nuclear waste."[12] In a memorandum to energy secretary Steven Chu[13], President Obama said: "The commission should conduct a comprehensive review of policies for managing the back end of the nuclear fuel cycle, including all alternatives for the storage, processing, and disposal of civilian and defence used nuclear fuel and nuclear waste." The commission's interim report was in July 2011, and its final report was submitted to Congress in January 2012. [Back]

k. Nuclear Energy Institute, US State by State Used Fuel and Payments to the Nuclear Waste Fund February 2017. The 35th state is Idaho, storing used fuel from Three Mile Island. [Back]

l. In 2004, Exelon reached agreement with the US Justice Department on recovering up to $300 million in storage costs for its used fuel to 2010. The agreement covered all of Exelon's 17 nuclear reactors, and the cash came from tax monies, not the Nuclear Waste Fund. In 2006, the US Federal Court awarded $143 million in damages to three related New England utilities and $40 million to the Sacramento Utility District for the same reason – the former had had to build dry storage facilities. Then $43 million was awarded to Pacific Gas & Electric. In 2007, Duke Energy negotiated $56 million on same basis for three plants, plus ongoing costs. Then Xcel Energy was awarded $116 million for costs associated with three reactors from 1998 to 2004 and Entergy Arkansas was awarded $48.6 million for costs to 2006. Progress Energy was awarded $82.8 million in 2008. Xcel Energy was awarded about $200 million in mid-2011 in relation to Prairie Island and Monticello plants, half through to 2008, half 2009-13. Other utilities have been suing the federal government to achieve the same result and billions of dollars are involved. [Back]

m. Congress established a trust fund for waste management in 1982 under the Office of Civilian Radioactive Waste Management

(OCRWM), and utilities have supplied $21.2 billion (plus some $14 billion investment returns) to this Nuclear Waste Fund through a 0.1 cent/kWh levy towards final disposal. [Back]

n. PFS then offered the facility to the Department of Energy for use from 2008 pending Yucca Mountain repository opening, suggesting that it would be very much cheaper for DOE than leaving the used fuel at reactor sites. While fuel ownership was originally intended to remain with utility customers, the proposal to DOE was that it would take ownership at the reactor site (as was legally required by 1998) and be responsible for moving it to PFS, and ultimately to Yucca Mountain. [Back]

o. Class A low-level waste contains radionuclides with the lowest concentrations and the shortest half-lives; Classes B and C contain greater concentrations of radionuclides with longer half-lives. Class A LLW must be contained for up to 100 years, Class B waste for up to 300 years and Class C waste for up to 500 years. [Back]

References

1. Data from US Energy Information Administration (www.eia.doe.gov) [Back]
2. International Isotopes Inc. Announces Continued Progress Towards the Licensing and Construction of its Planned Uranium De-conversion and Fluorine Extraction Processing Facility, International Isotopes Inc. news release (3 August 2010) [Back]
3. USEC Anticipates Loan Guarantee Decision by Early August, USEC news release (6 July 2009); Department of Energy Denies USEC's Loan Guarantee Application, USEC news release (28 July 2009); Department of Energy and USEC Announce Decision to Delay USEC Loan Guarantee Application Final Review, Department of Energy press release (4 August 2009); USEC Reports $58.5 million net income for 2009, USEC news release (1 March 2010); USEC Updates Status of American Centrifuge Project, USEC news release (4 May 2010, 3 Aug 2010) [Back]
4. Fast Reactor Technology: A Path to Long-Term Energy Sustainability, American Nuclear Society Position Statement (November 2005) [Back]
5. Green focus in US energy budget, World Nuclear News (8 May 2009) [Back]
6. Fatal blow to GNEP?, World Nuclear News (29 June 2009); Federal Register, Notices, Vol. 74, No. 123, pages 31017-31018 (29 June 2009) [Back]
7. Program on Technology Innovation: Room at the Mountain,

Electric Power Research Institute, Product ID: 1015046 (29 June 2007) [Back]
8. Yucca Mountain 'terminated', World Nuclear News (8 May 2009); FY 2010 Congressional Budget Request: U.S. Department of Energy Environmental Management/Defense Nuclear Waste Disposal/Nuclear Waste Disposal, DOE/CF-039, Volume 5, Office of Chief Financial Officer, Office of Budget (May 2009) [Back]
9. Ward Sproat: Comments on the DOE's Yucca Mountain Project, *Nuclear News*, Volume 50, Number 1 (January 2007) [Back]
10. Crossroads for Yucca Mountain, World Nuclear News (10 December 2008) [Back]
11. Yucca Mountain cost estimate rises to $96 billion, World Nuclear News (6 August 2008) [Back]
12. Secretary Chu Announces Blue Ribbon Commission on America's Nuclear Future, U.S. Department of Energy, Press Release (29 January 2010) [Back]
13. Blue Ribbon Commission on America's Nuclear Future, Office of the Press Secretary, The White House (29 January 2010) [Back]
14. Summary of Program Financial & Budget Information As of January 31, 2010, Office of Civilian Radioactive Waste Management, U.S. Department of Energy

General sources

American Centrifuge Plant page on the USEC website (www.usec.com)
Urenco USA (formerly National Enrichment Facility) page on the Urenco website (www.urenco.com)
Uranium Disposition Services website (www.uds-llc.com)
Assessment of Disposal Options for DOE-Managed High-Level Radioactive Waste and Spent Nuclear Fuel, October 2014, DOE.

Appendix: Shutdown Power Reactors 2019

Reactor Name	Type	MWt	Location	Shutdown	Status
Big Rock Point	BWR	240	Charlevoix, MI	29/08/97	DECON completed – ISFSI only
Bonus	BWR	50	Punta Higuera, PR	1/06/68	ENTOMB
Crystal River	PWR	2568	Florida	5/2/13	DECON by 2027
CVTR	PHWR	65	Parr, SC	1/01/67	SAFSTOR

Dresden 1	BWR	700	Morris, IL	31/10/78	SAFSTOR
Elk River	BWR	58	Elk River, MN	1/02/68	DECON completed
Fermi 1	FNR	200	Newport, MI	22/09/72	SAFSTOR
Fort Calhoun	PWR	c1428	Blair, NE	24/10/2016	DECON by mid-2020s
Fort St. Vrain	HTR	842	Platteville, CO	18/08/89	DECON completed, ISFSI only
Connecticut Yankee	PWR	1,825	Haddam Neck, CT	5/12/96	DECON completed, ISFSI only
Hallam	SCGMR	256	Hallam, NE	1/09/64	ENTOMB
Humboldt Bay	BWR	200	Eureka, CA	2/07/76	DECON in progress
Indian Point 1	PWR	615	Buchanan, NY	31/10/74	DECON by mid-2020s
Kewaunee	PWR	1772	Carlton, WI	7/5/13	SAFSTOR
LaCrosse	BWR	165	Genoa, WI	30/04/87	DECON completed
Maine Yankee	PWR	2,700	Wiscasset, ME	6/12/96	DECON completed - ISFSI only
Millstone 1	BWR	2,011	Waterford, CT	21/07/98	SAFSTOR
NS Savannah	PWR	74	Baltimore, MD	1/11/70	SAFSTOR
Oyster Creek	BWR	619	Ocean County, NJ	9/17/2018	DECON by 2030
Pathfinder	BWR	190	Sioux Falls, SD	16/09/67	DECON Completed
Peach Bottom 1	HTR	115	Peach Bottom, PA	31/10/74	SAFSTOR
Pilgrim 1	BWR	677	Plymouth, MA	31/05/2019	DECON
Piqua	OCMR	46	Piqua, OH	1/01/66	ENTOMB
Rancho Seco	PWR	2,772	Herald, CA	7/06/89	DECON completed – ISFSI only
San Onofre 1	PWR	1,347	San Clemente, CA	30/11/92	DECON completed – ISFSI only
San Onofre 2	PWR	3,438	San Clemente, CA	7/6/13	DECON by 2030
San Onofre 3	PWR	3,438	San Clemente, CA	7/6/13	DECON by 2030
Saxton	PWR	24	Saxton, PA	1/05/72	DECON completed

Shippingport	PWR	236	Shippingport, PA	1/01/82	DECON completed
Shoreham	BWR	2,436	Wading River, NY	28/06/89	DECON completed
Sturgis FNPP	PWR	10	Fort Belvoir, VA	1976	SAFSTOR
Three Mile Island 1	PWR	2770	Middletown, PA	2019	SAFSTOR
Three Mile Island 2	PWR	2,770	Middletown, PA	28/03/79	SAFSTOR: post defuelling monitored storage
Trojan	PWR	3,411	Ranier, OR	9/11/92	DECON completed – ISFSI only
Vallecitos (GE VBWR)	BWR	50	Pleasanton, CA	9/12/63	SAFSTOR
Vermont Yankee	BWR	1593	Vernon, VT	29/12/14	DECON by 2030
Yankee Rowe	PWR	600	Franklin Co., MA	1/10/91	DECON completed – ISFSI only
Zion 1	PWR	3,250	Zion, IL	21/02/97	DECON by 2021
Zion 2	PWR	3,250	Zion, IL	19/09/96	DECON by 2021
Total: 36 (to 2019)					

ISFSI = Independent spent fuel storage installation remaining under licence and regulation by NRC, usually allowing the rest of the site to be released.

See also NRC Backgrounder on Decommissioning Nuclear Power Plants

Related information

USA: Nuclear Power
USA: Nuclear Power Policy
International Framework for Nuclear Energy Cooperation
Military Warheads as a Source of Nuclear Fuel